

Mechatronics

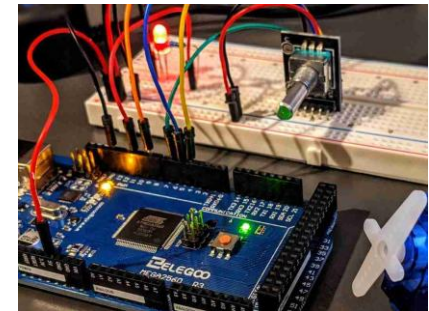
Mechatronics (MEE4033) · Department of Mechanical Engineering

Lecture 7

Circuit Basic Experiments

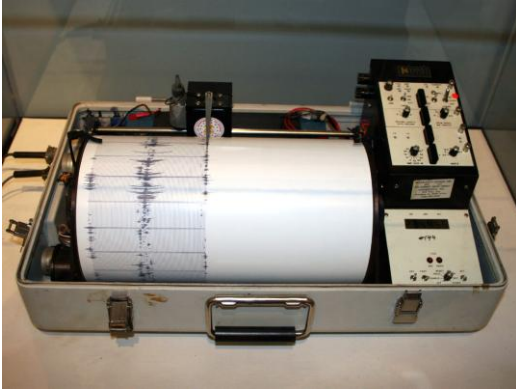
Principles and Applications of Electrical Engineering, 7th ed., Chapter 4

Laboratory reference: NI myDAQ and NI ELVISmx



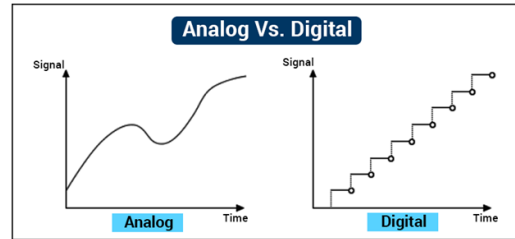
Digital Signal Acquisition

Analog: seismometer



[Seismometer reference](#)

- Low flexibility
- Continuous measurement in real time
- Can be used in analog devices only
- Large volume and high weight
- Observational errors



Digital: DAQ board



[NI data acquisition systems](#)

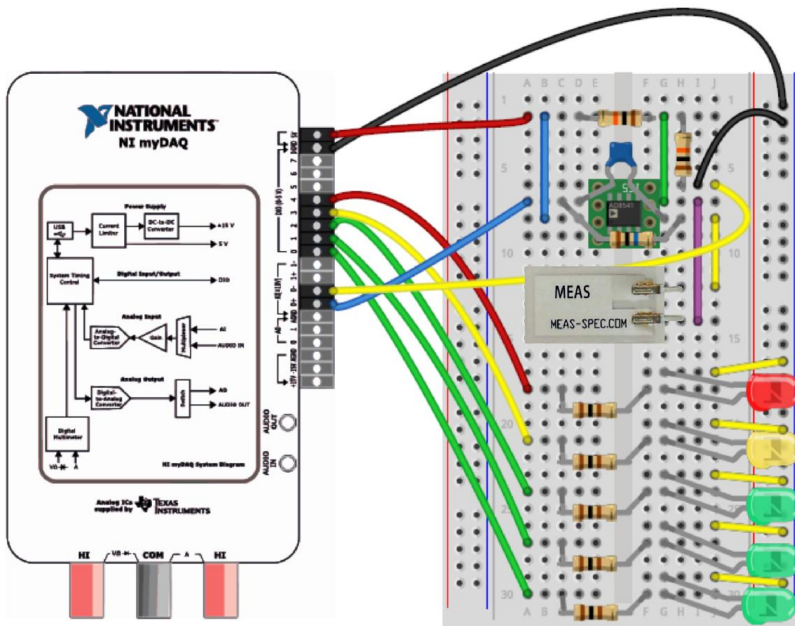
- High flexibility
- Discrete measurements every T_s
- Requires high computational power
- Compact
- Delay
- Aliasing
- Approximation errors

myDAQ: Interfaces and Resources



NI myDAQ

- Analog input: 2 differential channels
- Analog audio input: 1
- Analog output: 2 channels
- Digital I/O: 8 lines
- Supply outputs: ± 15 V and +5 V



Documents and video tutorials

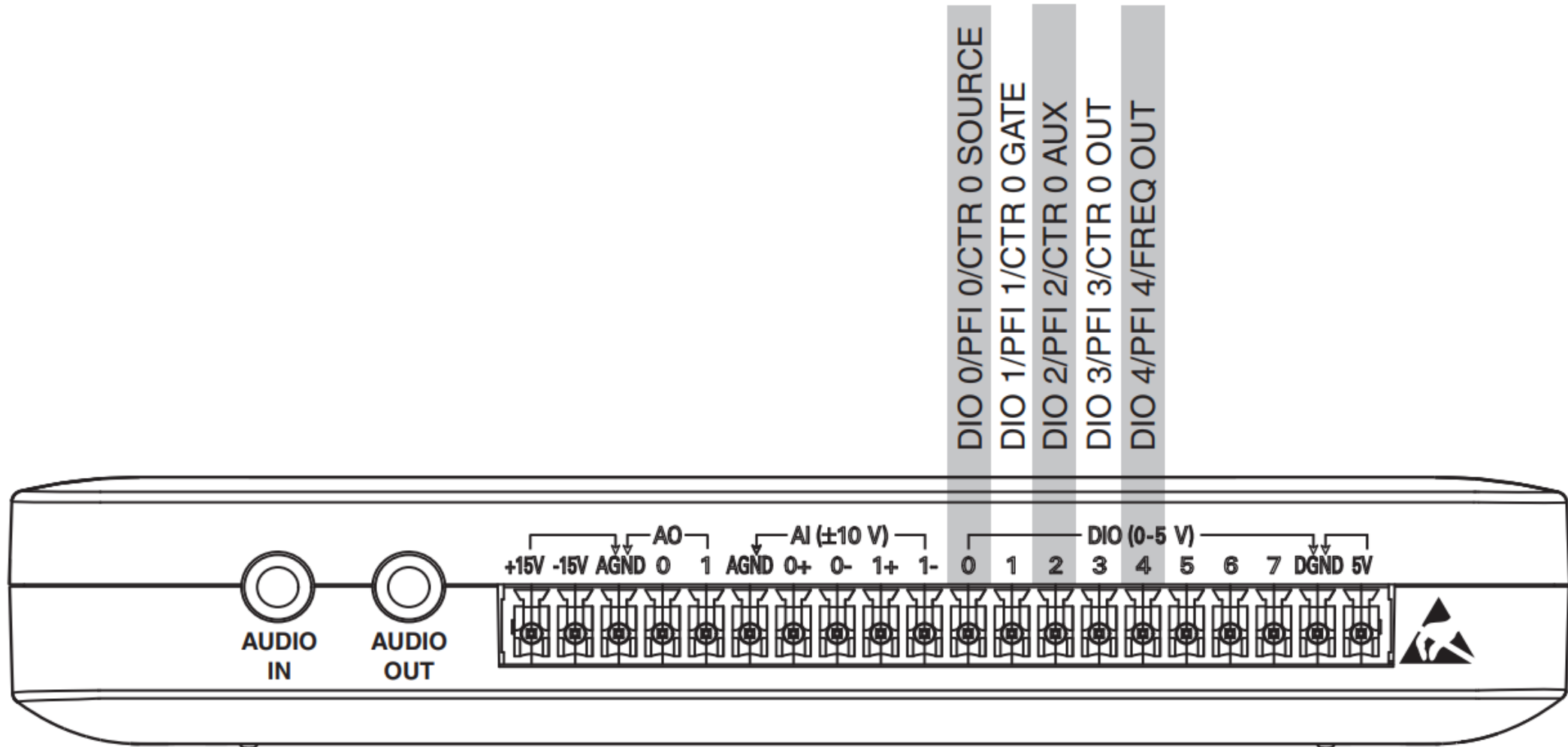
[NI myDAQ specifications](#)

[NI myDAQ user guide](#)

[YouTube tutorial 1](#)

[YouTube tutorial 2](#)

myDAQ: Terminal Layout



myDAQ Screw Terminal Signals (1)

Table 1. Screw Terminal Signal Descriptions

Signal Name	Reference	Direction	Description
AUDIO IN	—	Input	Audio Input —Left and right audio inputs on a stereo connector
AUDIO OUT	—	Output	Audio Output —Left and right audio outputs on a stereo connector
+15V/-15V	AGND	Output	+15 V/-15 V power supplies
AGND	—	—	Analog Ground —Reference terminal for AI, AO, +15 V, and -15 V
AO 0/AO 1	AGND	Output	Analog Output Channels 0 and 1*
AI 0+/AI 0-; AI 1+/AI 1-	AGND	Input	Analog Input Channels 0 and 1

myDAQ Screw Terminal Signals (2)

Table 1. Screw Terminal Signal Descriptions (Continued)

Signal Name	Reference	Direction	Description
DIO <0..7>	DGND	Input or Output	Digital I/O Signals —General-purpose digital lines or counter signals
DGND	—	—	Digital Ground —Reference for the DIO lines and the +5 V supply
PFI 0/ CTR 0 SOURCE	—	—	Digital I/O, line 0; PFI 0, Default function: Counter 0 Source
PFI 1/ CTR 0 GATE	—	—	Digital I/O, line 1; PFI 1, Default function: Counter 0 Gate
PFI 2/ CTR 0 AUX	—	—	Digital I/O, line 2; PFI 2, Default function: Counter 0 Aux
PFI 3/ CTR 0 OUT	—	—	Digital I/O, line 3; PFI 3, Default function: Counter 0 Out
PFI 4/ FREQ OUT	—	—	Digital I/O, line 4; PFI 4, Default function: Frequency Output
5V	DGND	Output	5 V power supply
* The myDAQ supports output on either a ± 10 V or ± 2 V range and output from either one or both AO channels. When outputting from both AO channels, however, they must share a common voltage range.			

Laboratory Software

MATLAB most recent version

Download and install the full package (recommended).

[Download page](#)

Arduino IDE

Install Arduino IDE.

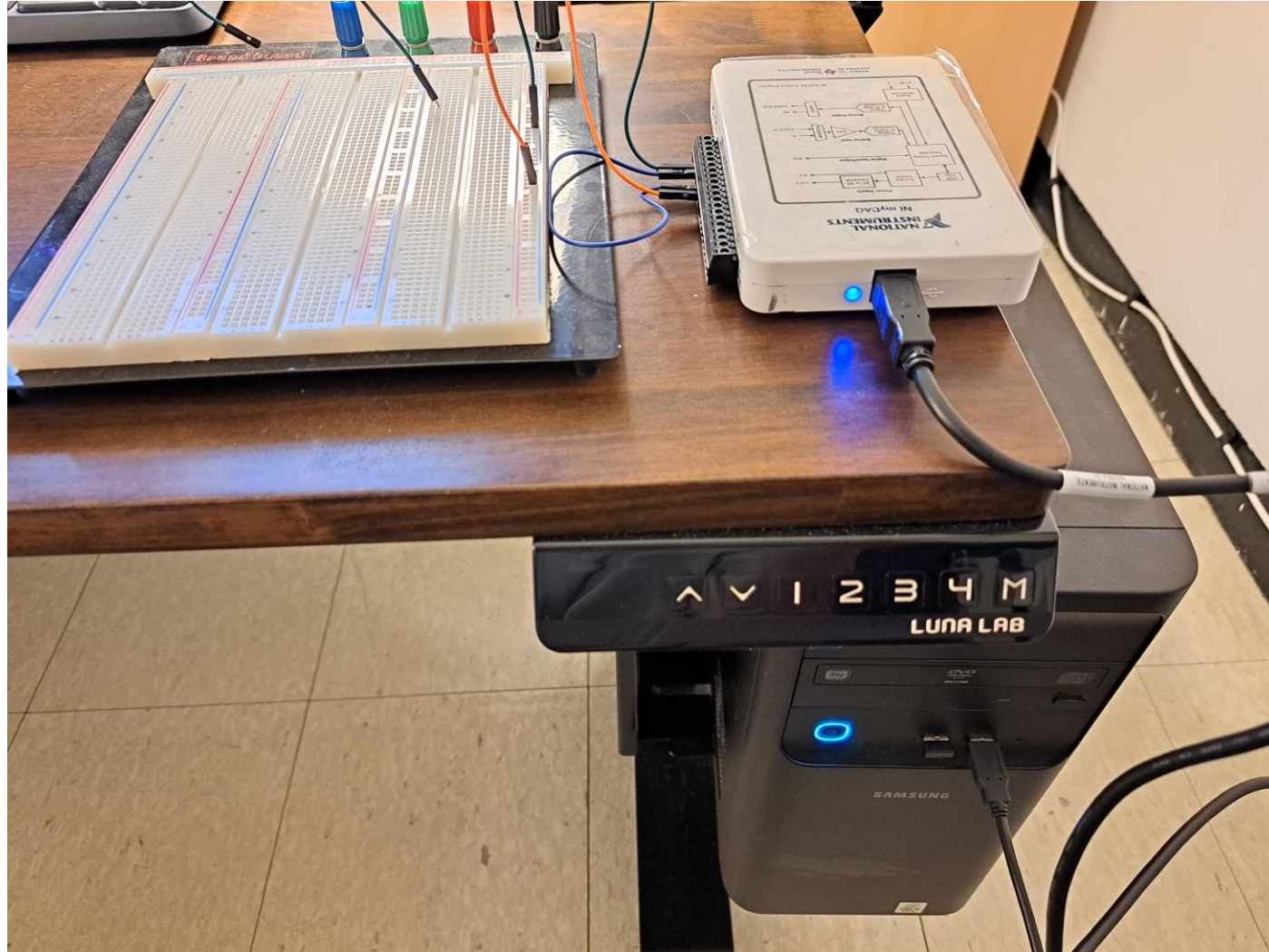
[Download page](#)

NI ELVISmx 19.0

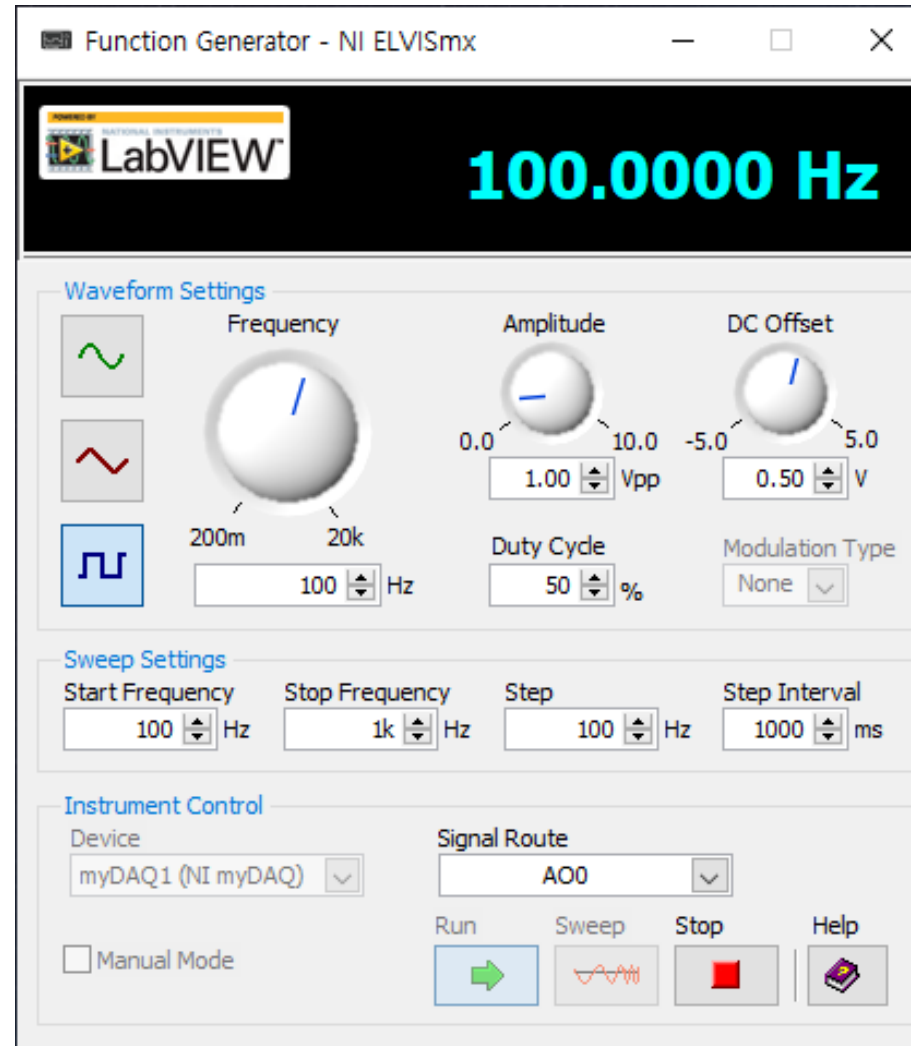
Install the virtual-instrument software.

[Download page](#)

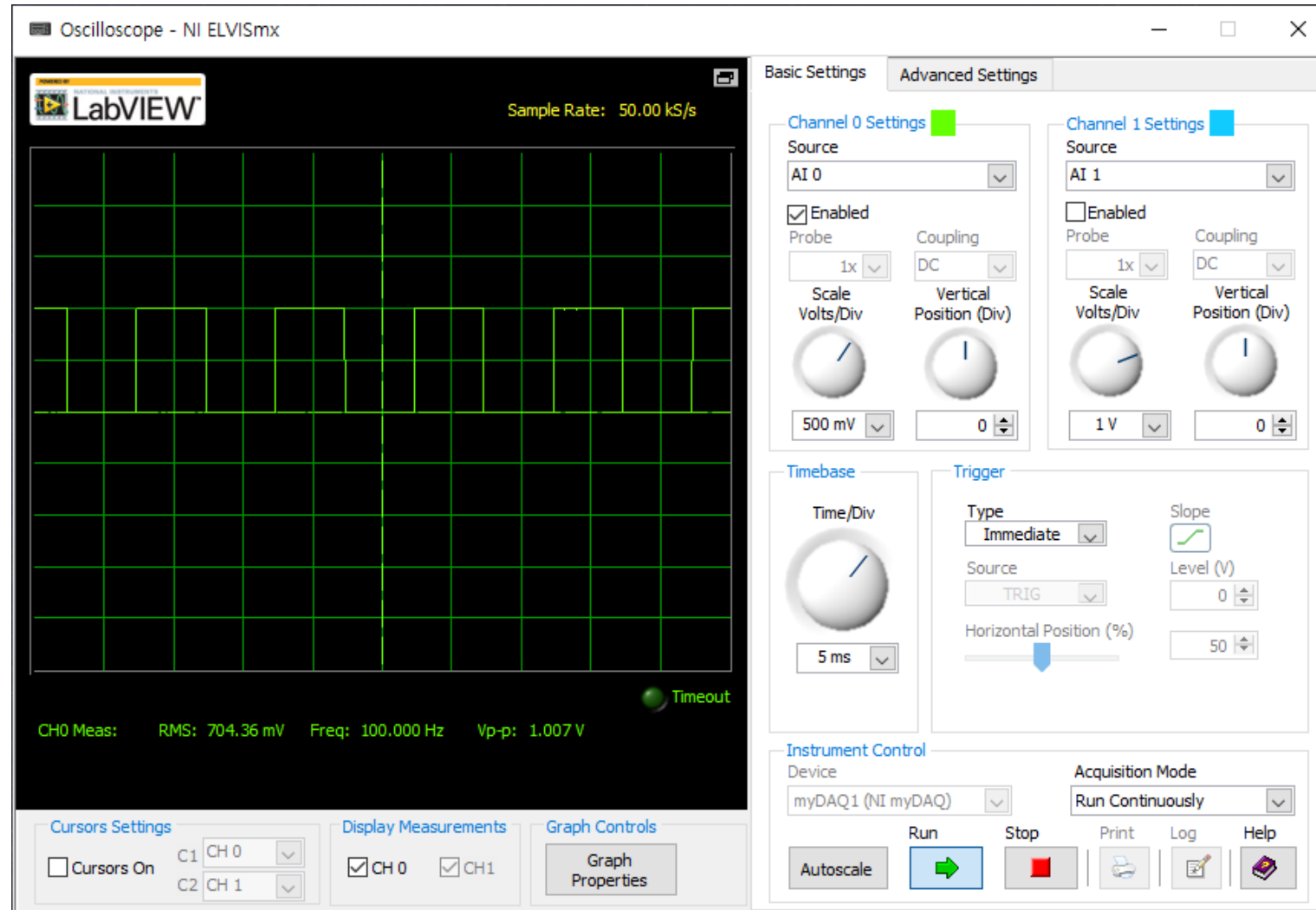
myDAQ: Laboratory Setup



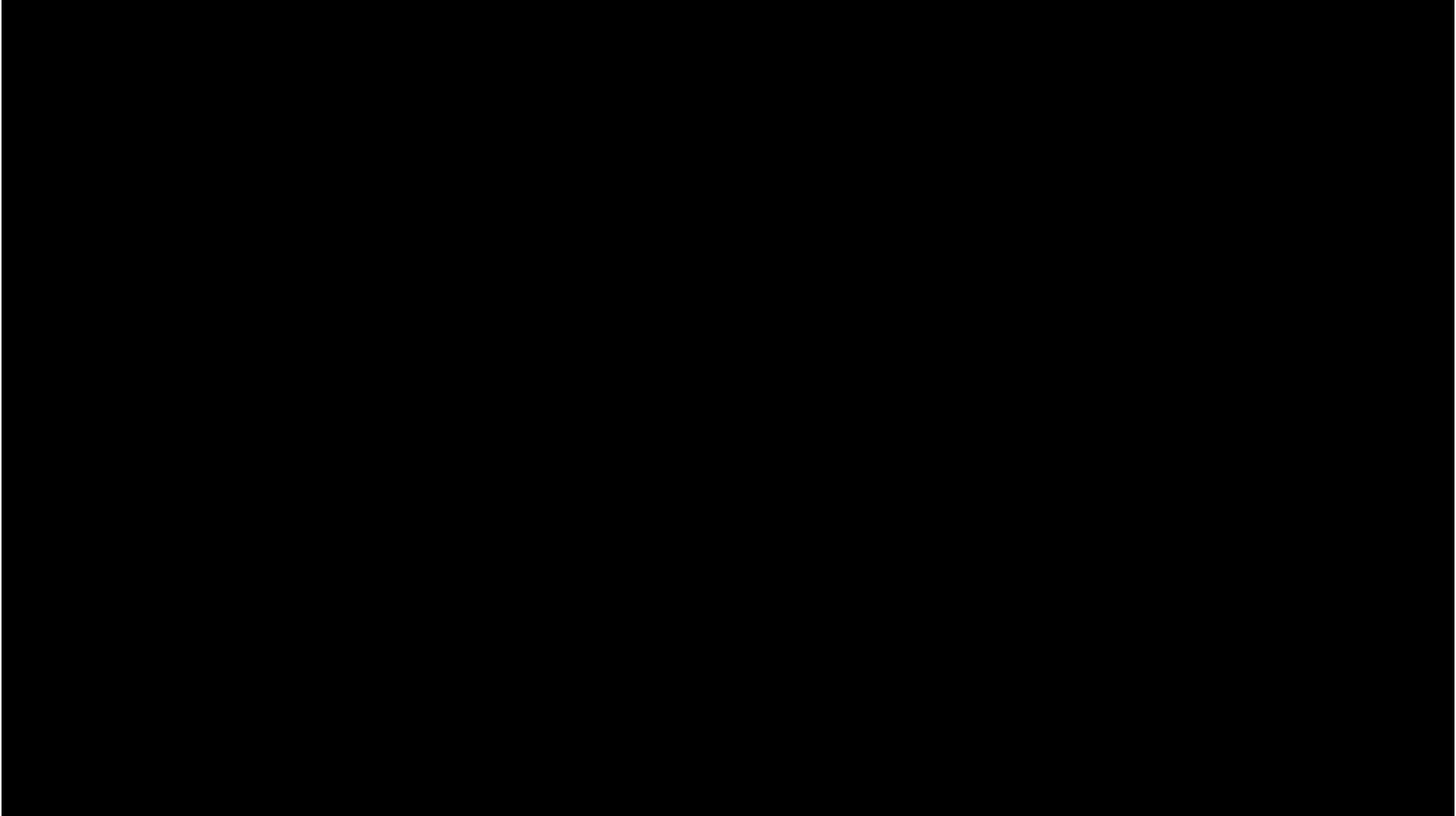
NI ELVISmx: Function Generator



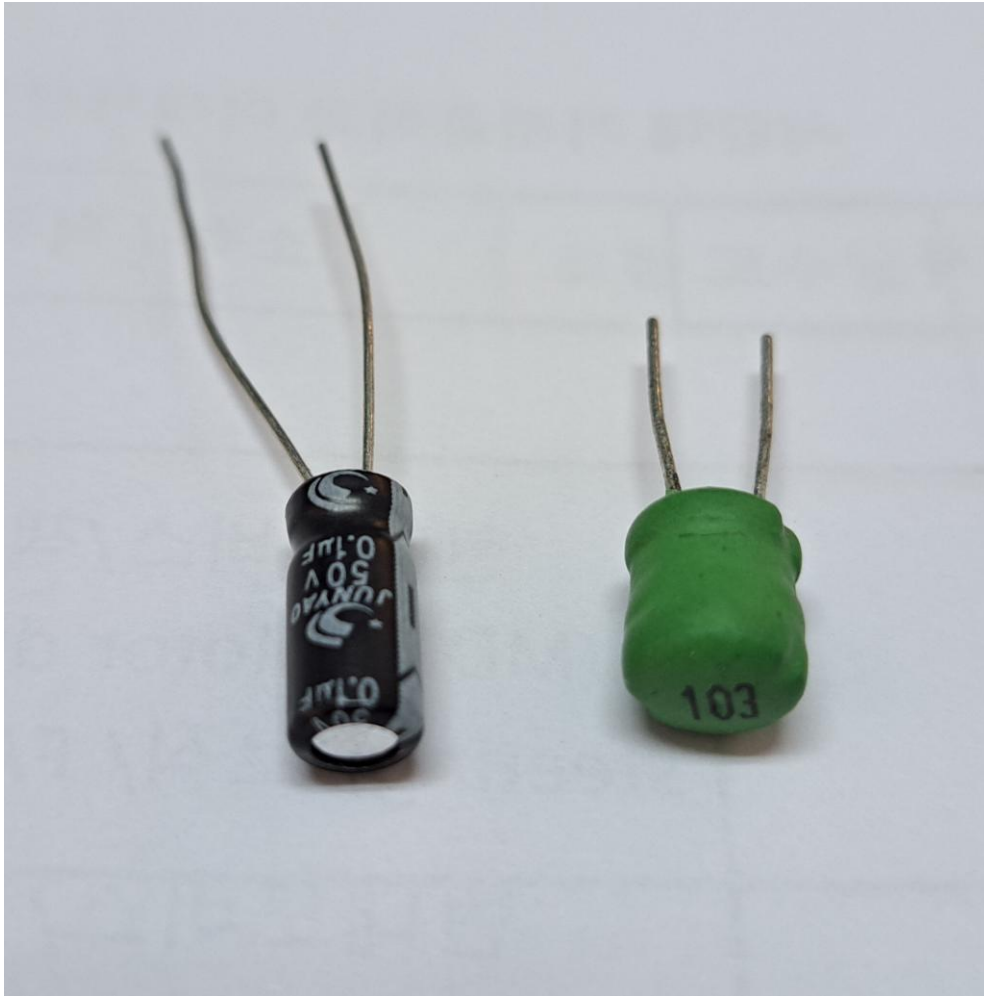
NI ELVISmx: Oscilloscope



NI ELVISmx: Demonstration



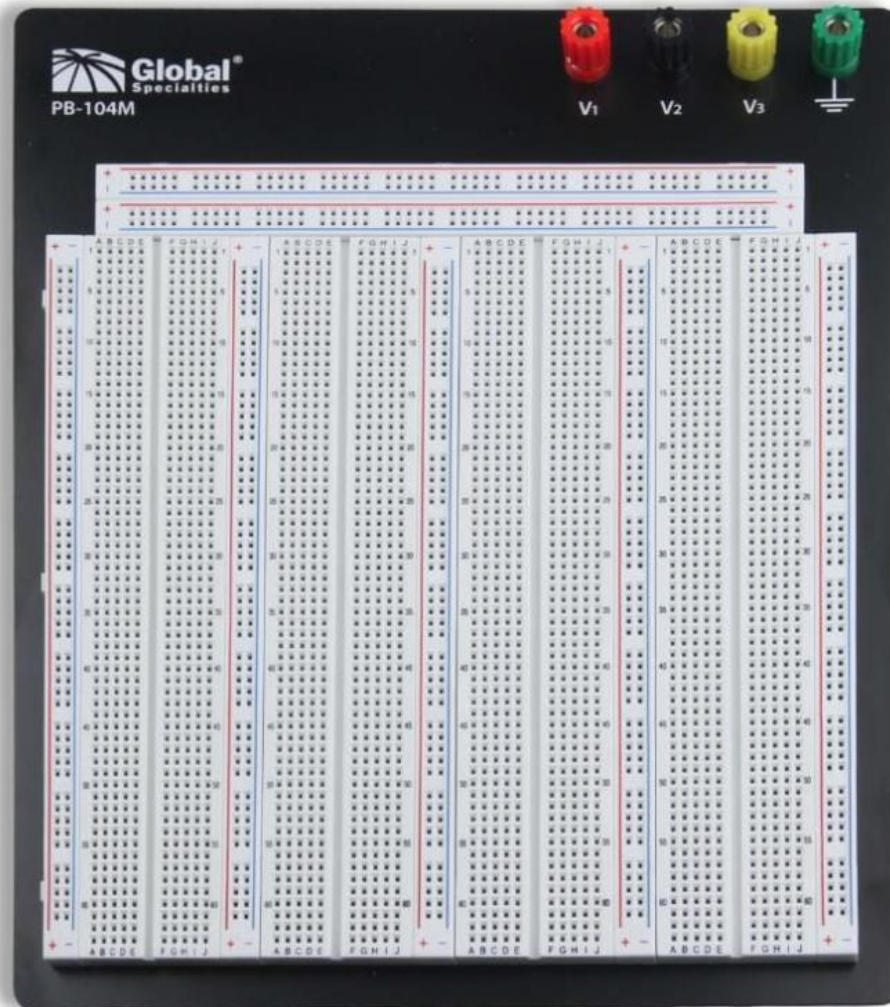
Circuit Components



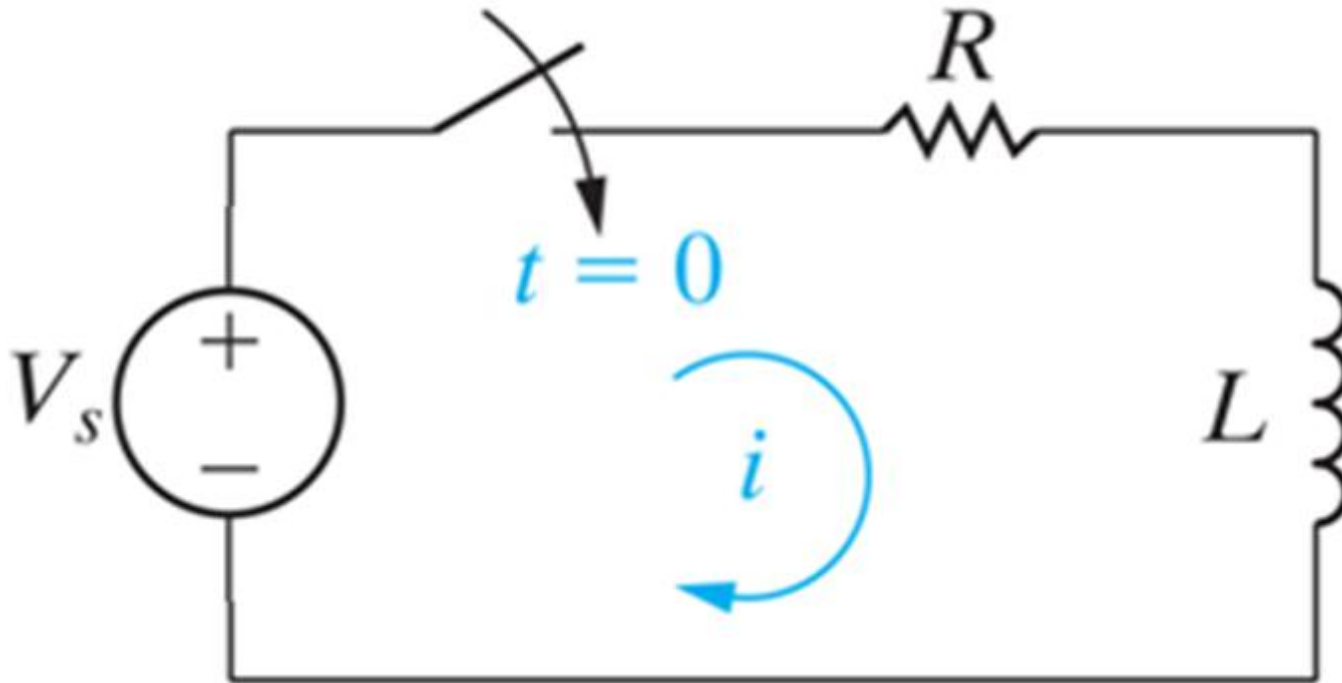
$$C = 0.1 \mu\text{F}$$

$$L = 10 \text{ mH}$$

Breadboard



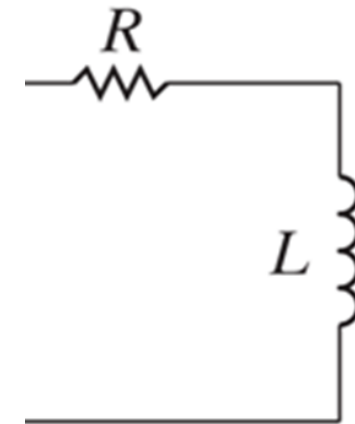
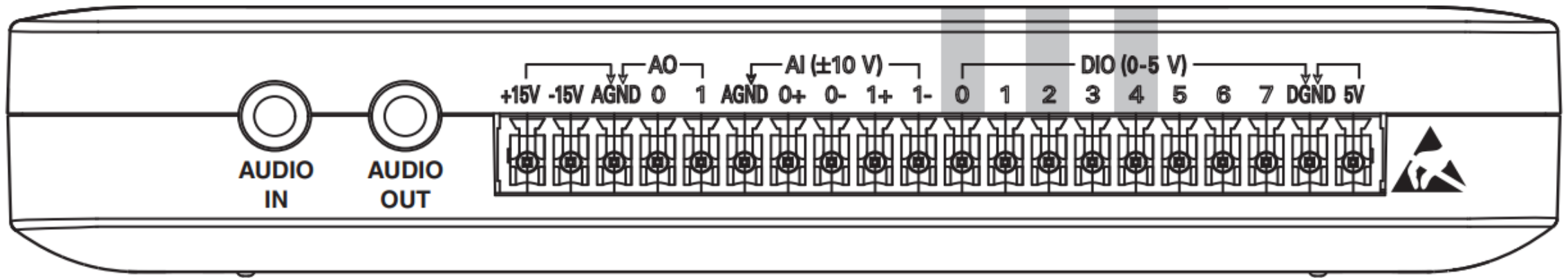
RL Circuit: Step Response



Series RL circuit

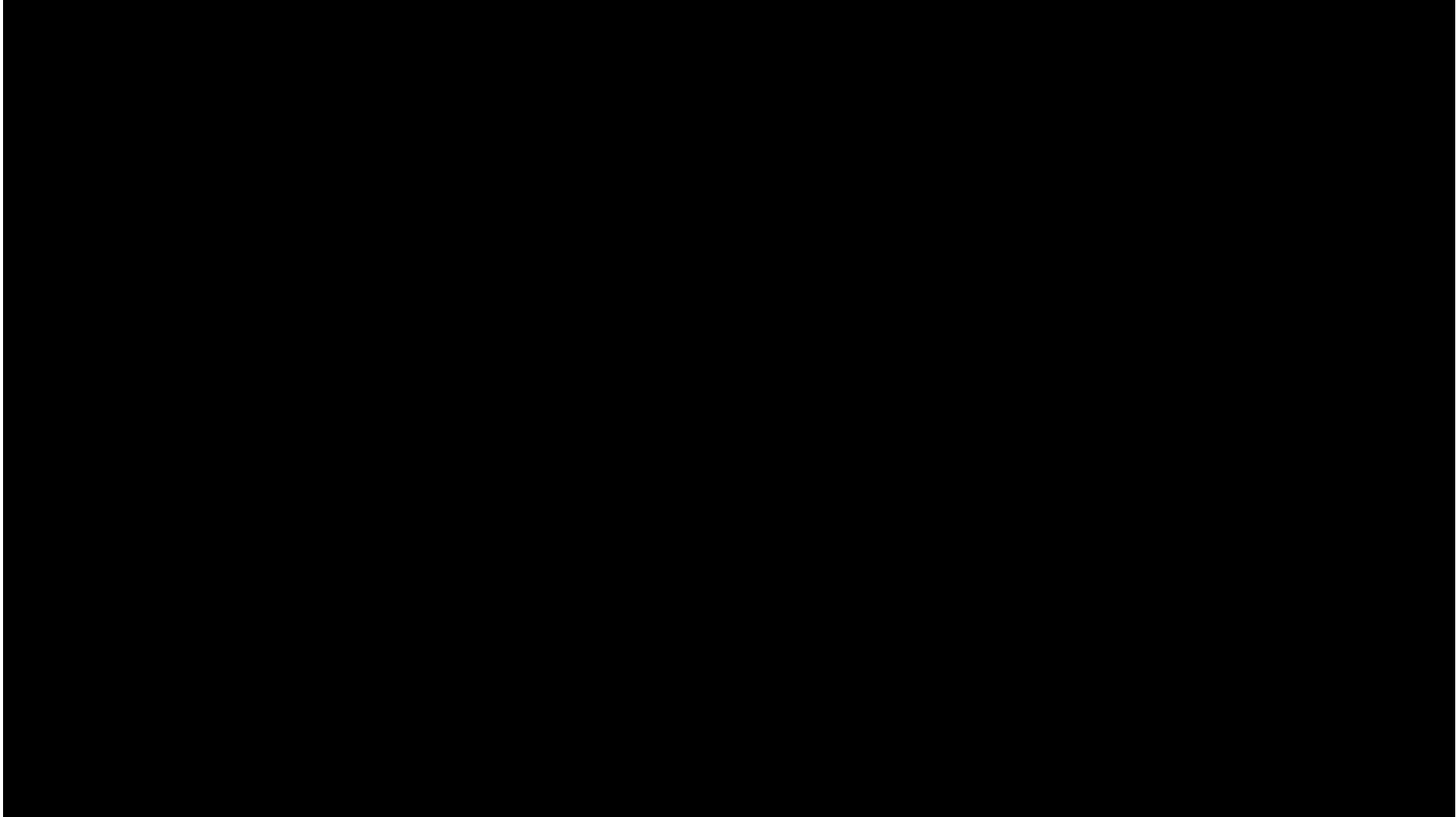
- $L = 20 \text{ mH}$, $R = 470 \Omega$
- $L = 20 \text{ mH}$, $R = 1 \text{ k}\Omega$
- $L = 20 \text{ mH}$, $R = 2.2 \text{ k}\Omega$

RL Circuit: myDAQ Connections

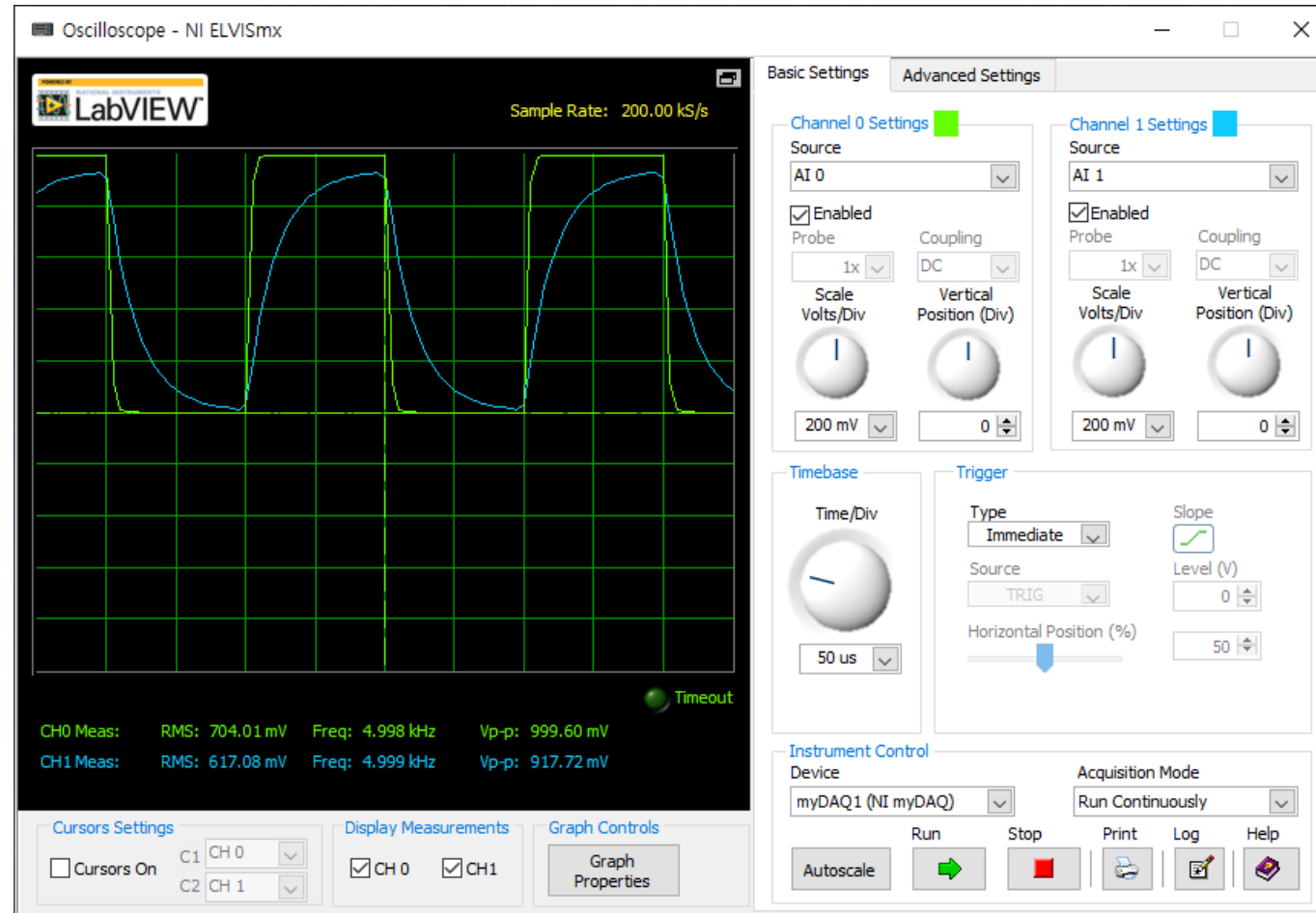




RL Circuit: Measurement Demonstration

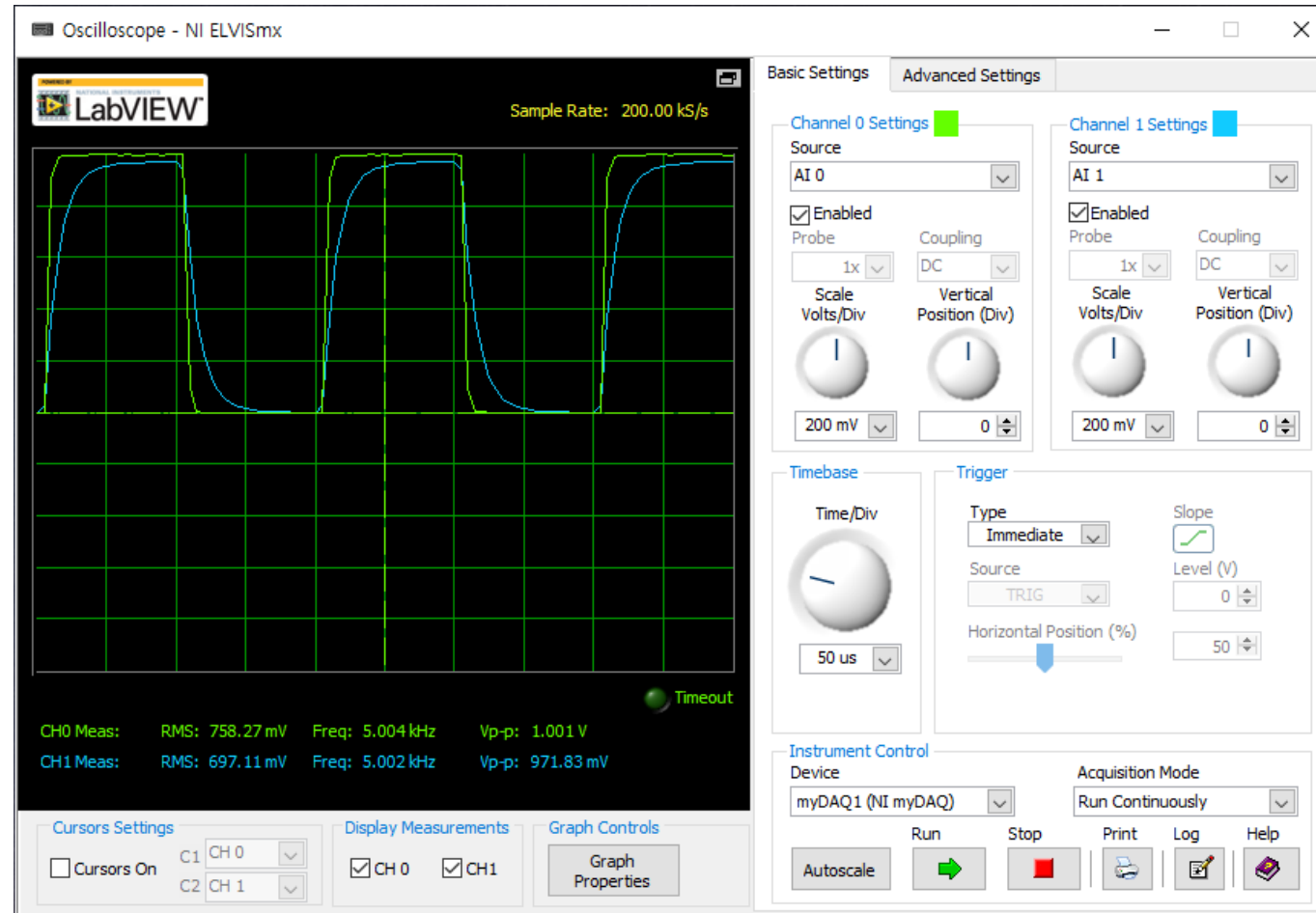


RL Circuit: Measured Response, $R = 470\ \Omega$



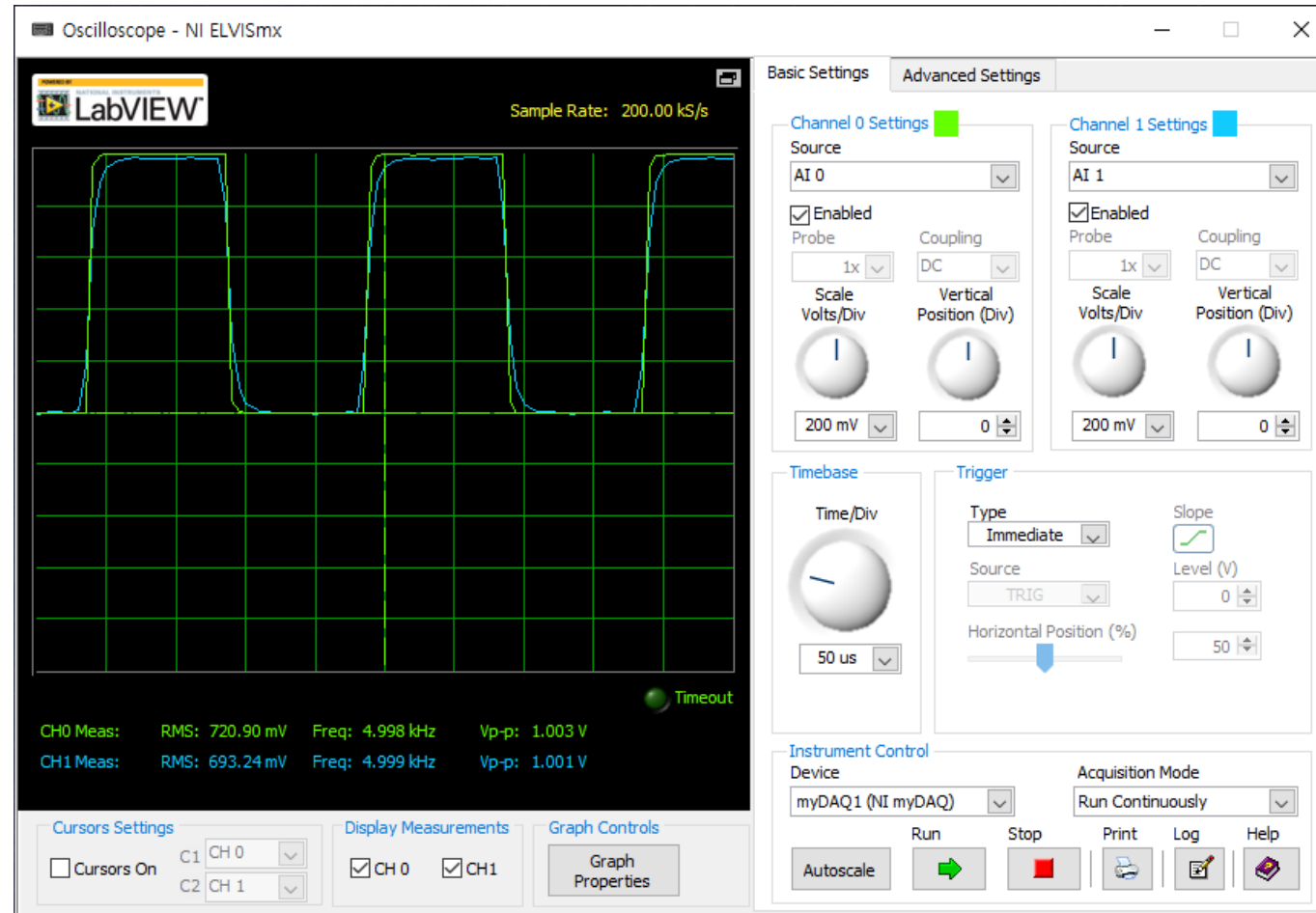
$$L = 10\text{ mH}, R = 470\ \Omega$$

RL Circuit: Measured Response, $R = 1\text{ k}\Omega$



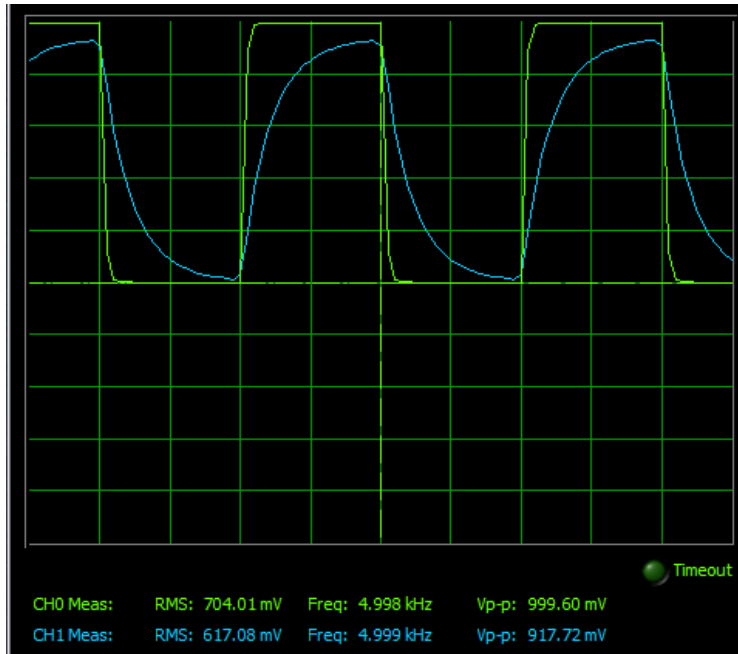
$L = 10\text{ mH}$, $R = 1\text{ k}\Omega$

RL Circuit: Measured Response, $R = 2.2 \text{ k}\Omega$

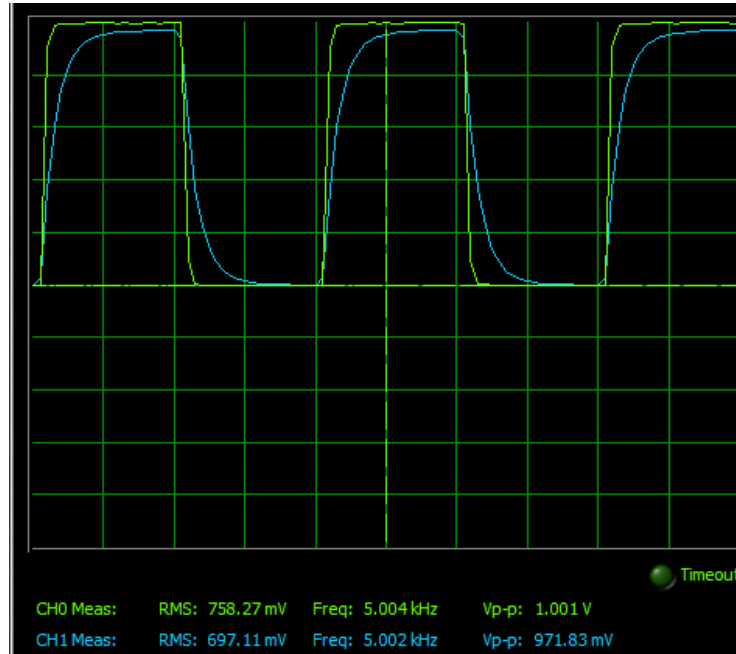


$L = 10 \text{ mH}$, $R = 2.2 \text{ k}\Omega$

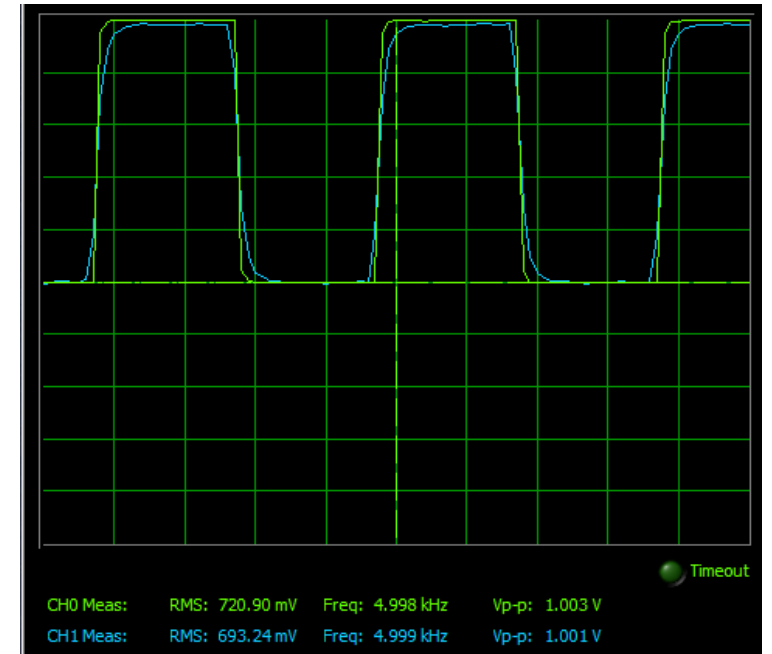
RL Circuit: Comparison of Three Resistances



$L = 10 \text{ mH}, R = 470 \Omega$

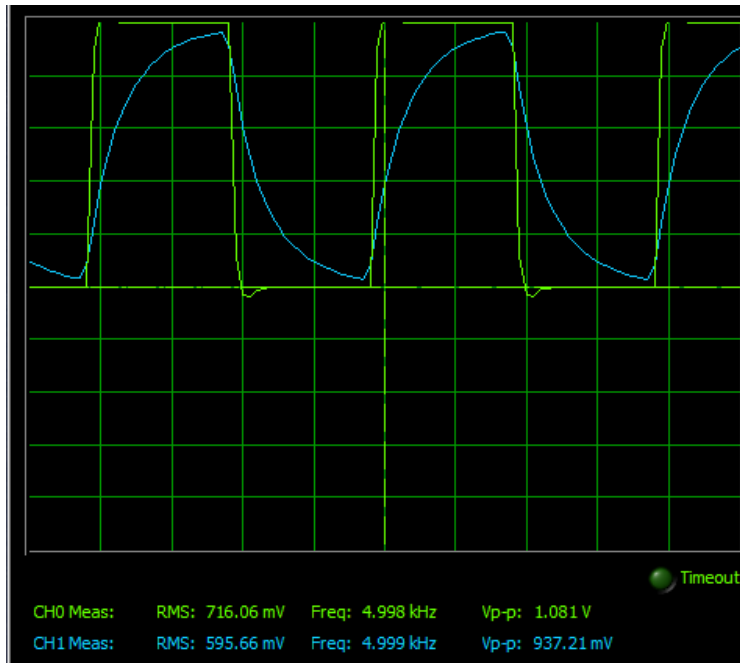


$L = 10 \text{ mH}, R = 1 \text{ k}\Omega$

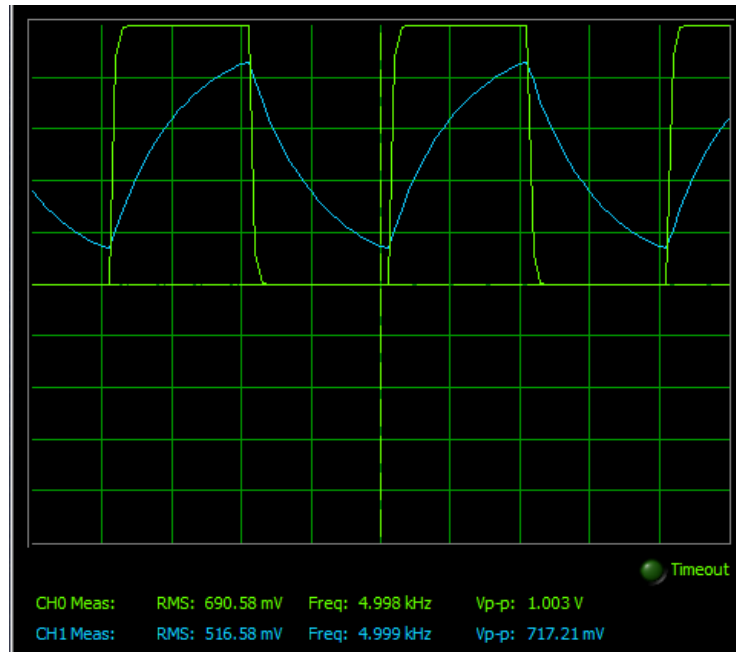


$L = 10 \text{ mH}, R = 2.2 \text{ k}\Omega$

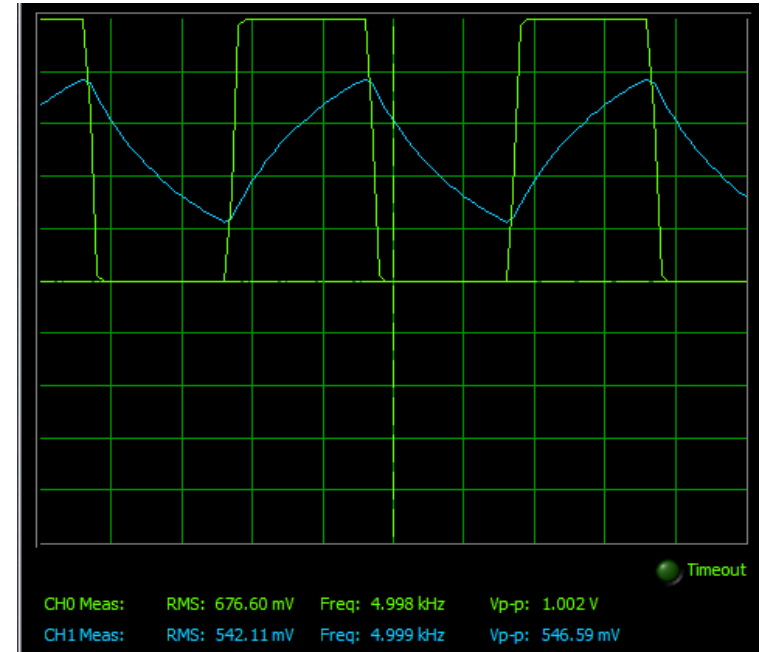
RC Circuit: Comparison of Three Resistances



$C = 0.1 \mu\text{F}$, $R = 330 \Omega$



$C = 0.1 \mu\text{F}$, $R = 680 \Omega$



$C = 0.1 \mu\text{F}$, $R = 1000 \Omega$

Homework

1. Investigate the working principles and pros/cons of single-ended ADC and differential ADC.

2. Series RL circuit

Measure the voltage across R.

$L = 10 \text{ mH}$

$R = 470 \, \Omega, 1 \text{ k}\Omega, 2.2 \text{ k}\Omega$

3. Series RC circuit

Measure the voltage across C.

$C = 0.1 \, \mu\text{F}$

$R = 330 \, \Omega, 680 \, \Omega, 1000 \, \Omega$

Input for both circuits: PWM, 5 kHz, 1 Vpp, +0.5 V DC offset

For each circuit and each resistance:

- Attach a photograph of the experimental setup.
- Capture the input PWM and measured voltage simultaneously on a v-t graph.
- Show the full derivation of the measured voltage as a function of time.
- Compare the time constants obtained from the measured graph and the derivation.